

Dear Editor(s) of Cell Press,

We are pleased to submit our manuscript entitled “Bifurcation Analysis of a Cancer-Immunity Cycle” for consideration through the Cell Press multi-journal submission system. We believe that this work is particularly well suited for Cell Reports Physical Science, but we would also welcome consideration for any other Cell Press journal that the editors deem appropriate.

In this manuscript, we develop a rigorous bifurcation and stability analysis of a discrete-time mathematical model describing the cancer-immunity cycle under immune suppression. By classifying equilibrium structures on the trace-determinant plane, we identify both saddle-node bifurcations and stability bifurcations that mark transitions between immune-limited and immune-escape tumor states. Because immune suppression is the primary bifurcation parameter in the model, the results are directly linked to PD-1-mediated immune evasion and thus provide a theoretical framework for understanding the action and limitations of anti-PD-1 therapies. In particular, we show that the bifurcation structure differs between strong and weak immune regimes: under strong immune conditions, loss of stability may precede the saddle-node bifurcation, implying that standard bifurcation criteria alone can overestimate the effectiveness of immunotherapy. These findings suggest that such modeling approaches may be useful for informing and personalizing immunotherapy strategies based on individual immune system characteristics.

We believe this work will be of interest to readers in mathematical oncology, computational and systems biology, and quantitative medicine, as it bridges dynamical systems theory with biologically and clinically motivated questions in tumor-immune interactions. Thank you very much for your consideration. We look forward to your response.

Sincerely,

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